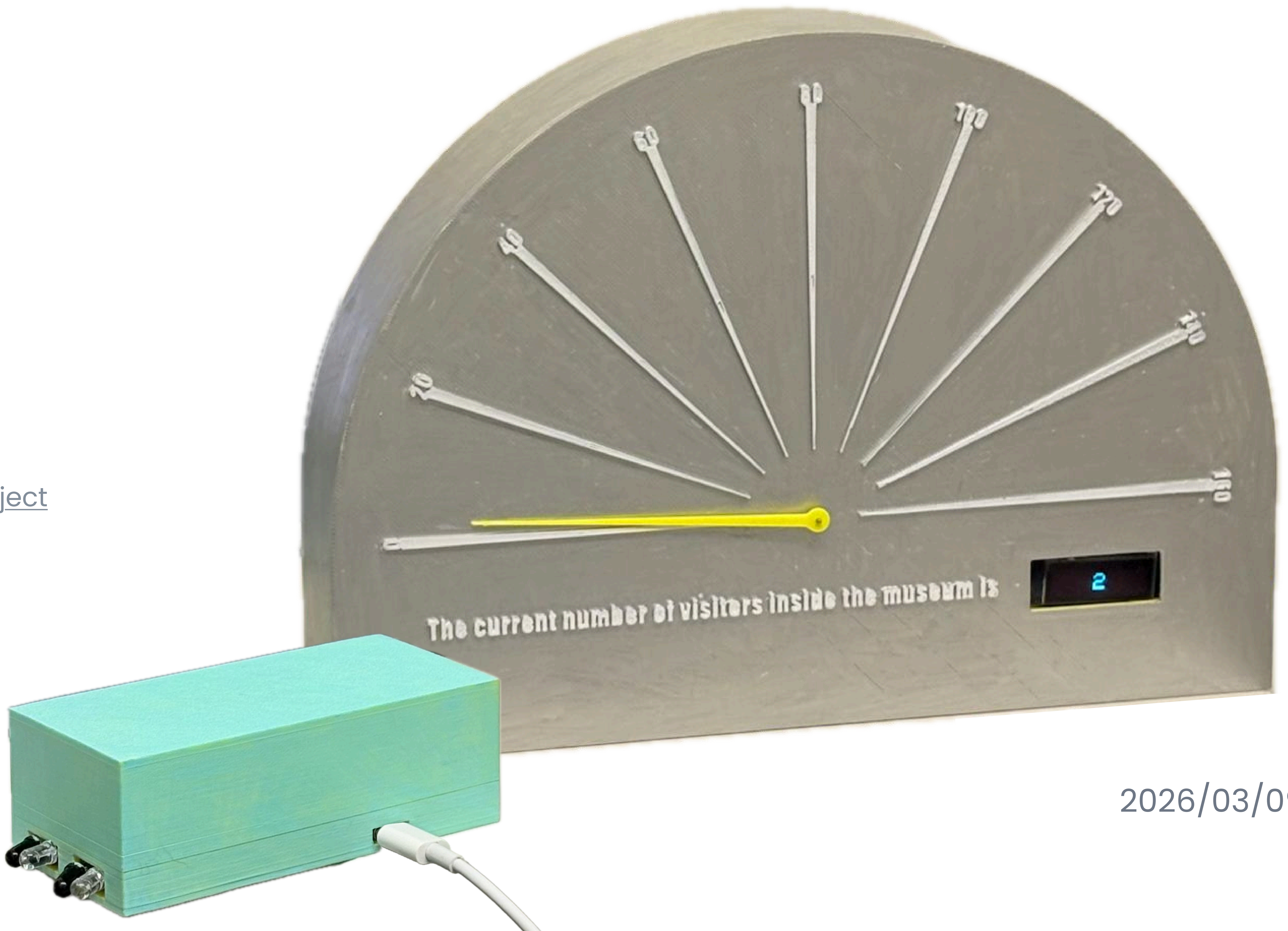
Museum Entrance Visitor Counter

Final Project Presentation

TECHIN 514A

GitHub repo: https://github.com/yuwennnnnnnnnnnn/514/tree/main/FinalProject

This project designed to **count visitors entering and exiting a museum space**, using sensors at the entrance and a physical display to **present real-time occupancy information.**



Yuwen Chen

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Problem being solved

Museums and galleries often rely on **staff to manually count** visitors at the entrance.

Manual counting is **inefficient** and often **inaccurate** during peak hours.

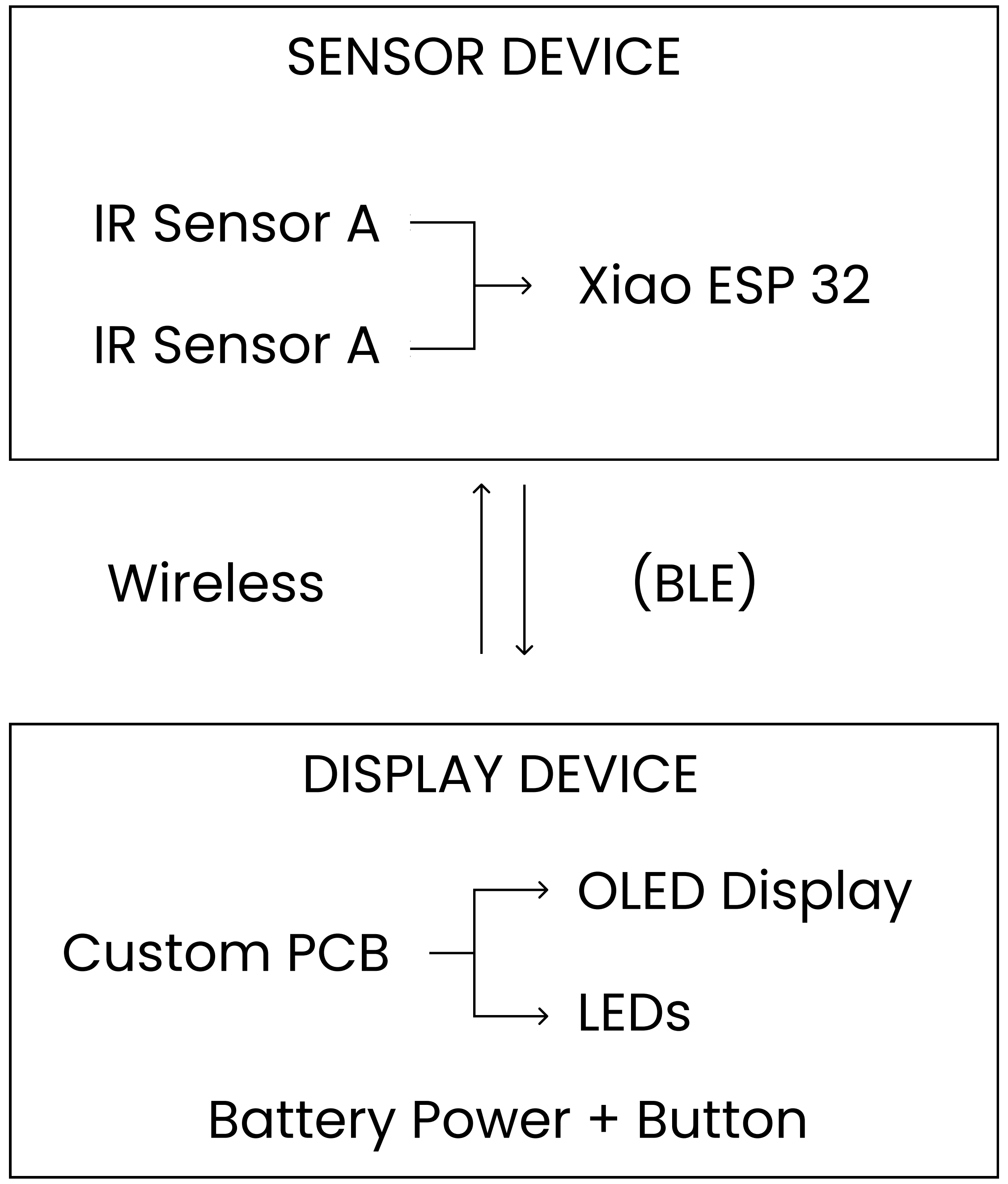
This project **automates visitor counting** and shows occupancy through a physical gauge display.

Proposed solution:

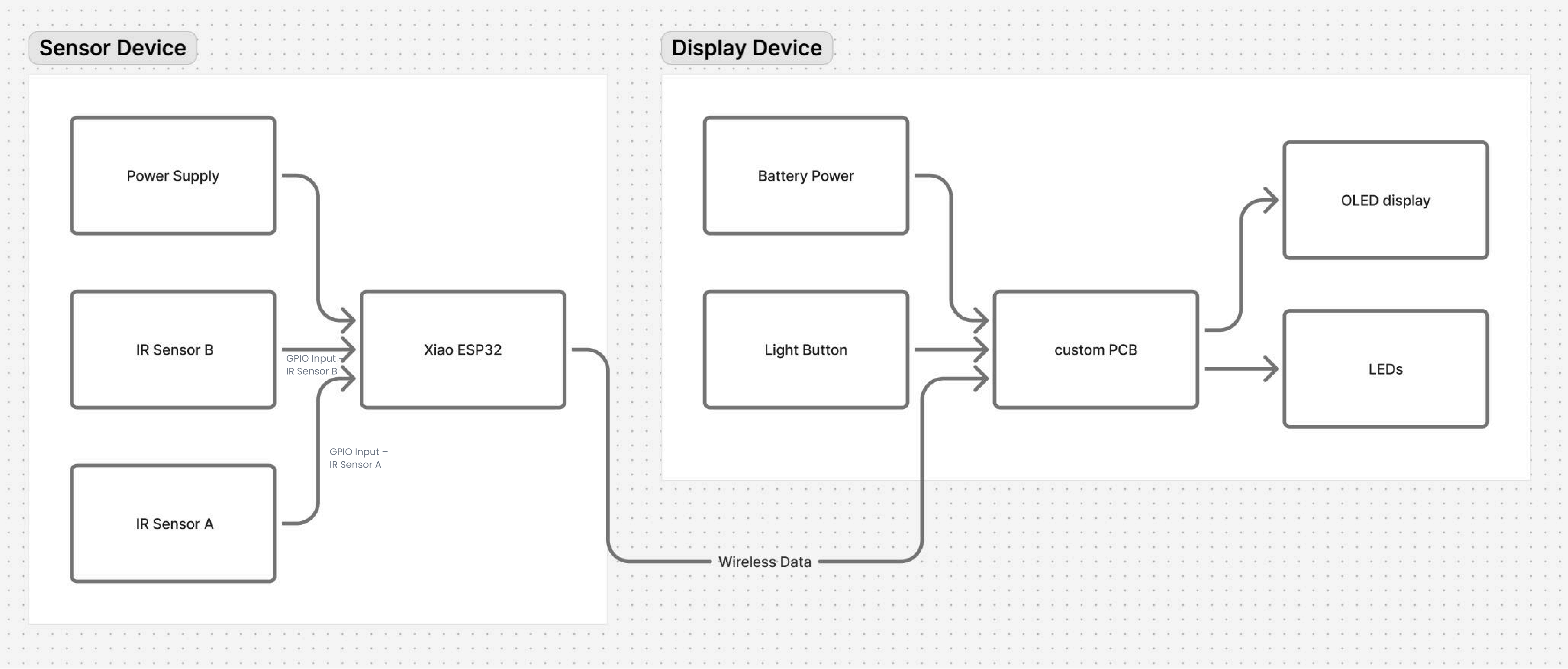
- **Sensing device:** A device installed at the museum entrance that uses distance sensors to detect people entering and exiting. The device processes directional movement and wirelessly sends visitor count data.
- **Display device:** physically displays the current visitor count or occupancy level, allowing staff to quickly understand space usage at a glance and adjust lighting and air conditioning accordingly.

System Structure & Device Communication

Device Communication Chart

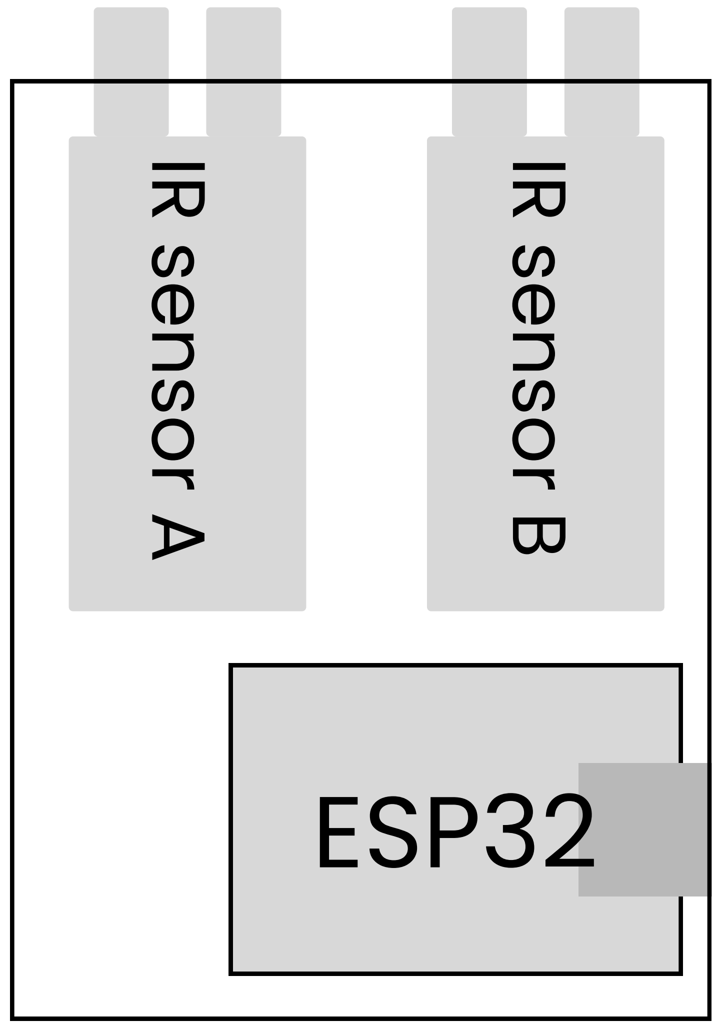


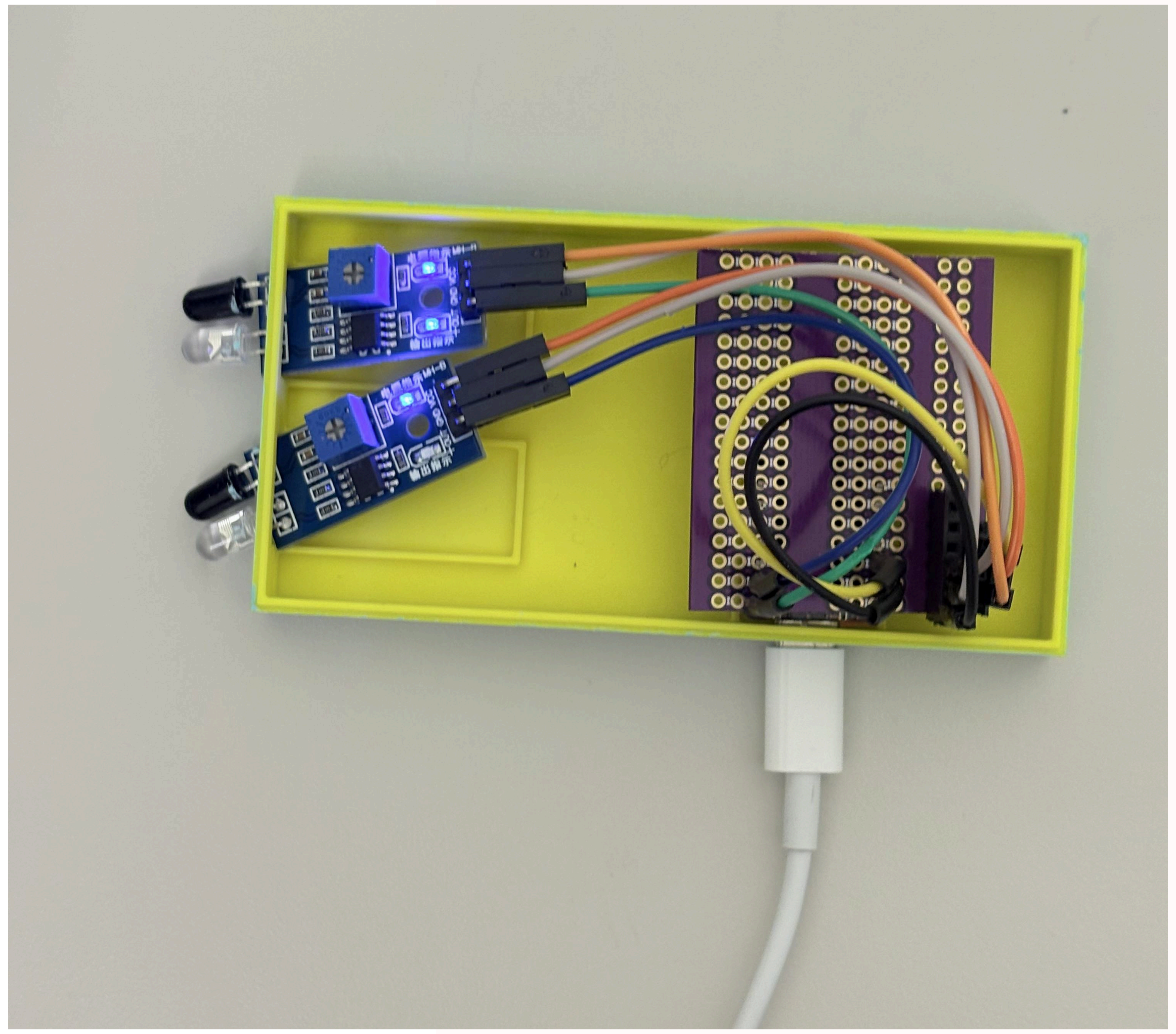
System Block Diagram

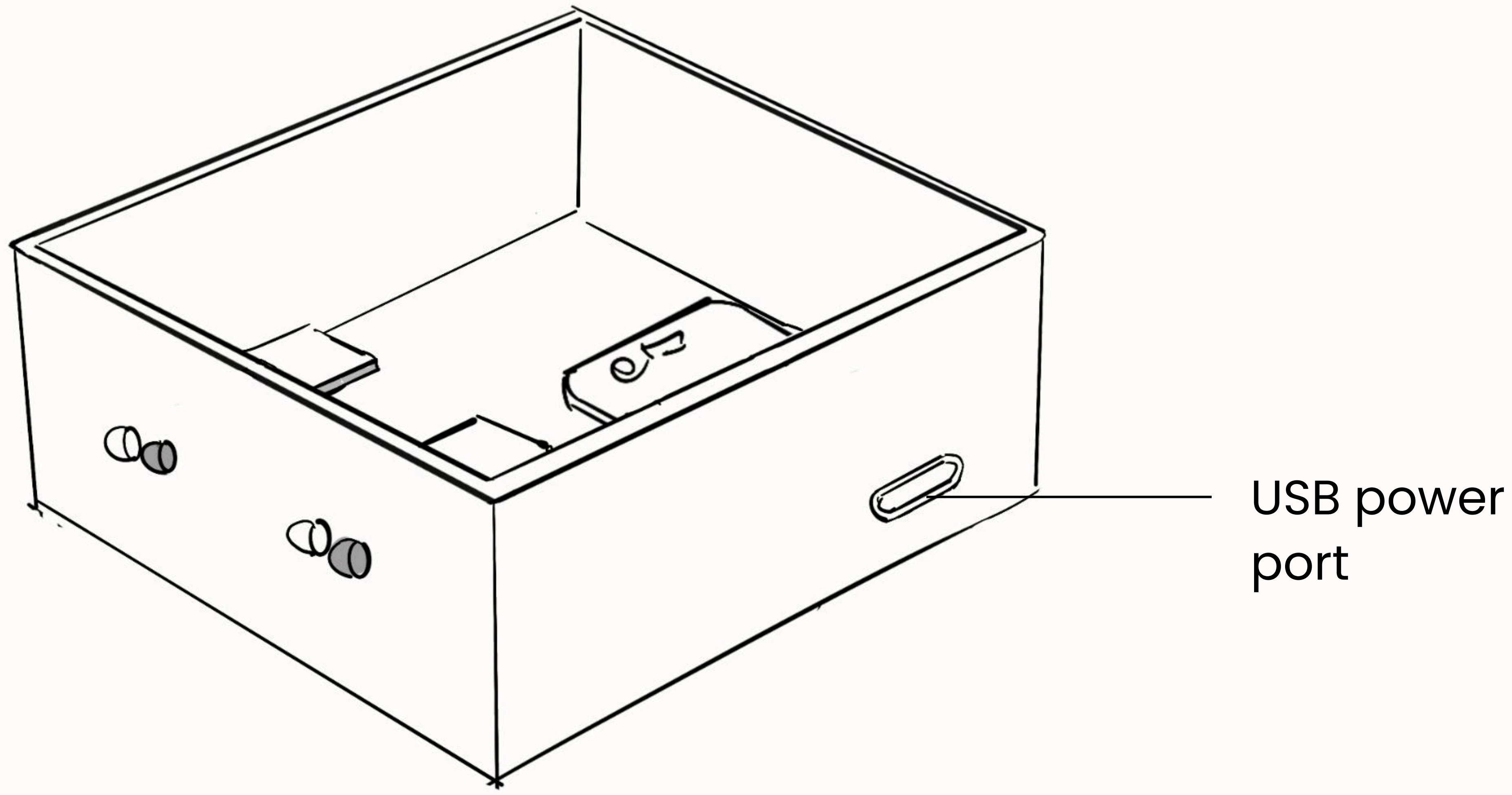


The sensor device acts as the data source, while the display device acts as the output interface.
Communication is handled wirelessly using Wi-Fi or Bluetooth.

Critical components of sensor device

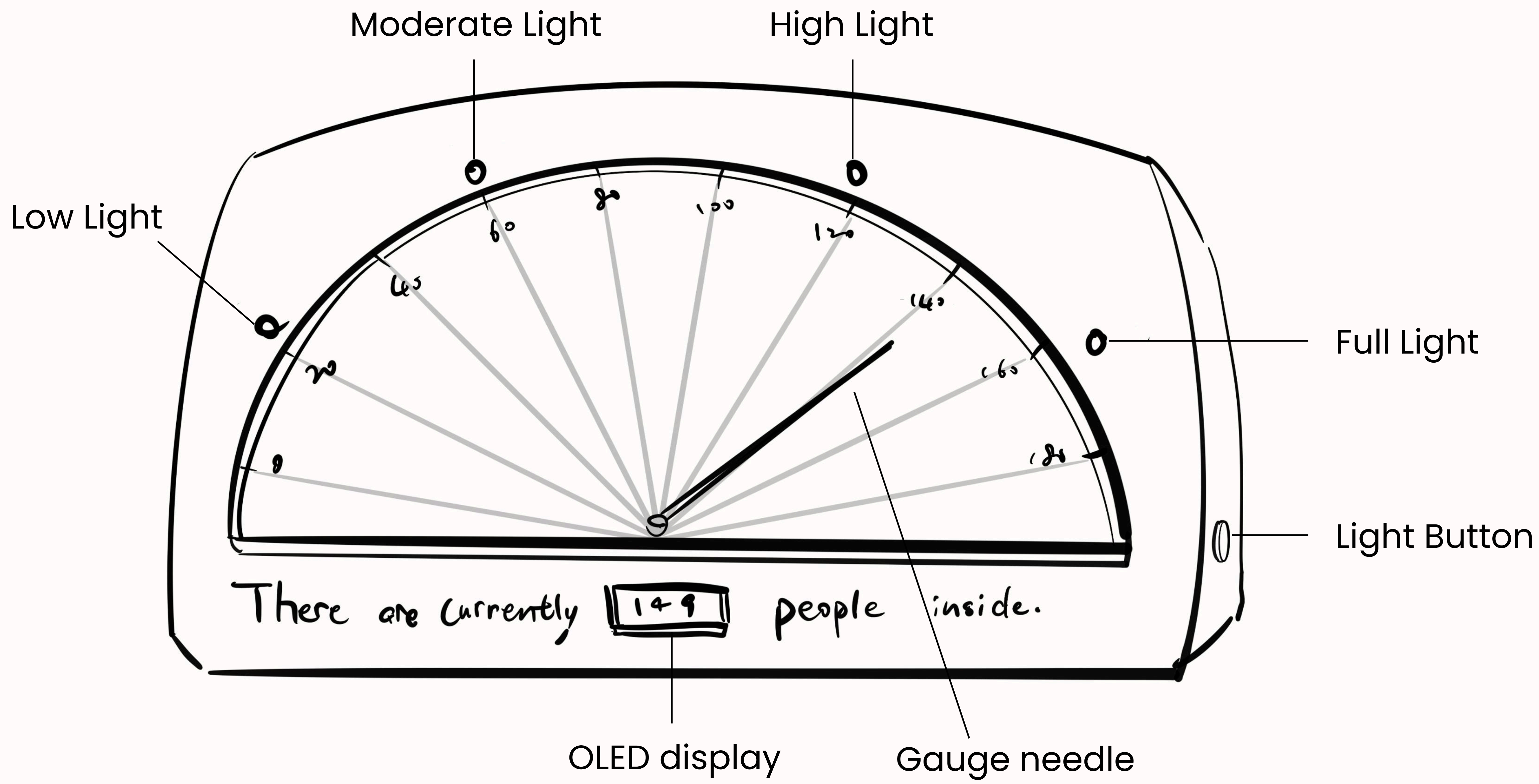


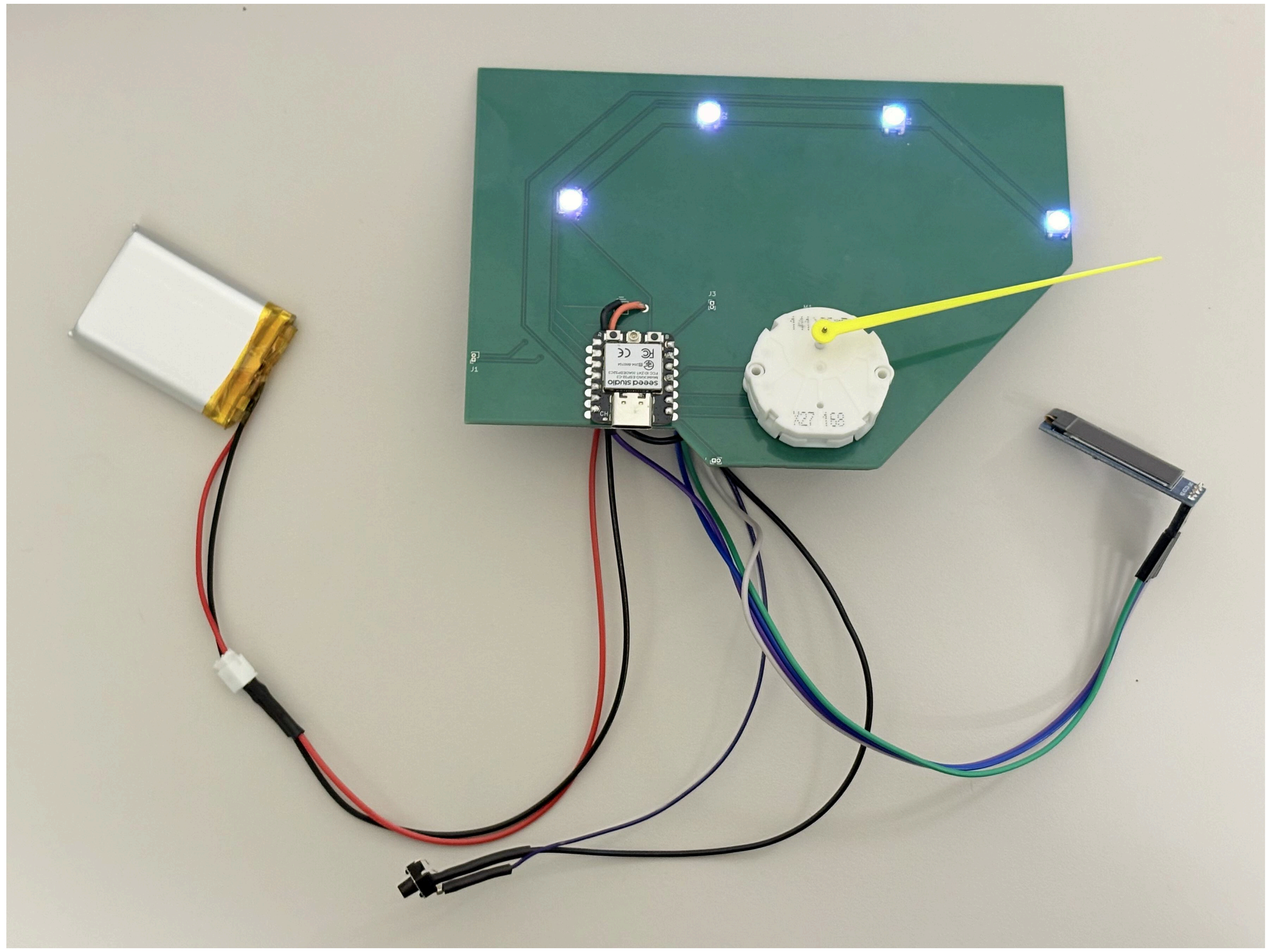




The sensor device uses two IR sensors to detect movement direction. When a person passes through the entrance, the order in which the sensors are triggered determines whether the count increases (entry) or decreases (exit). The microcontroller processes this logic and sends updated count data wirelessly.

Critical components of display device



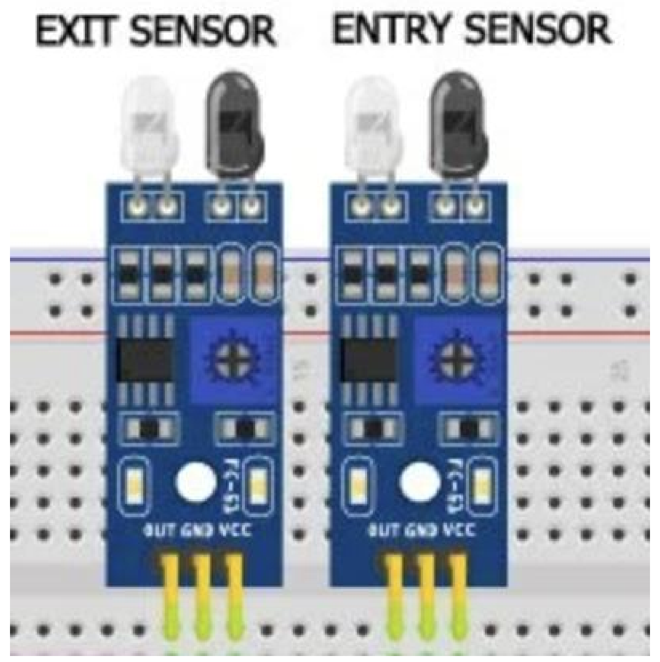


The display device receives visitor count data wirelessly from the sensor device. It shows the current number of visitors and translates that number into a simple visual status using LEDs and gauge, allowing staff to understand occupancy at a glance.

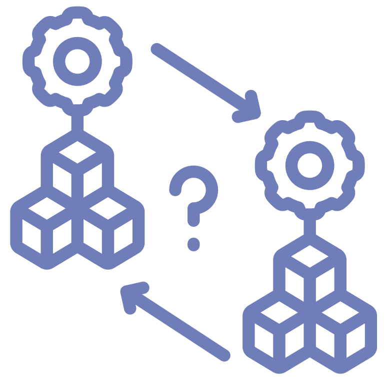
Signal processing

Visitor detection pipeline:

IR Sensors → Direction Detection → Noise Filtering → Count Update → BLE Transmission → OLED Display



The order of sensor triggers determines movement direction:
Sensor A → Sensor B = Entry (+1)
Sensor B → Sensor A = Exit (-1)



500ms debounce delay after each trigger to reject false double-counts

The ESP32 updates the visitor count in real time.

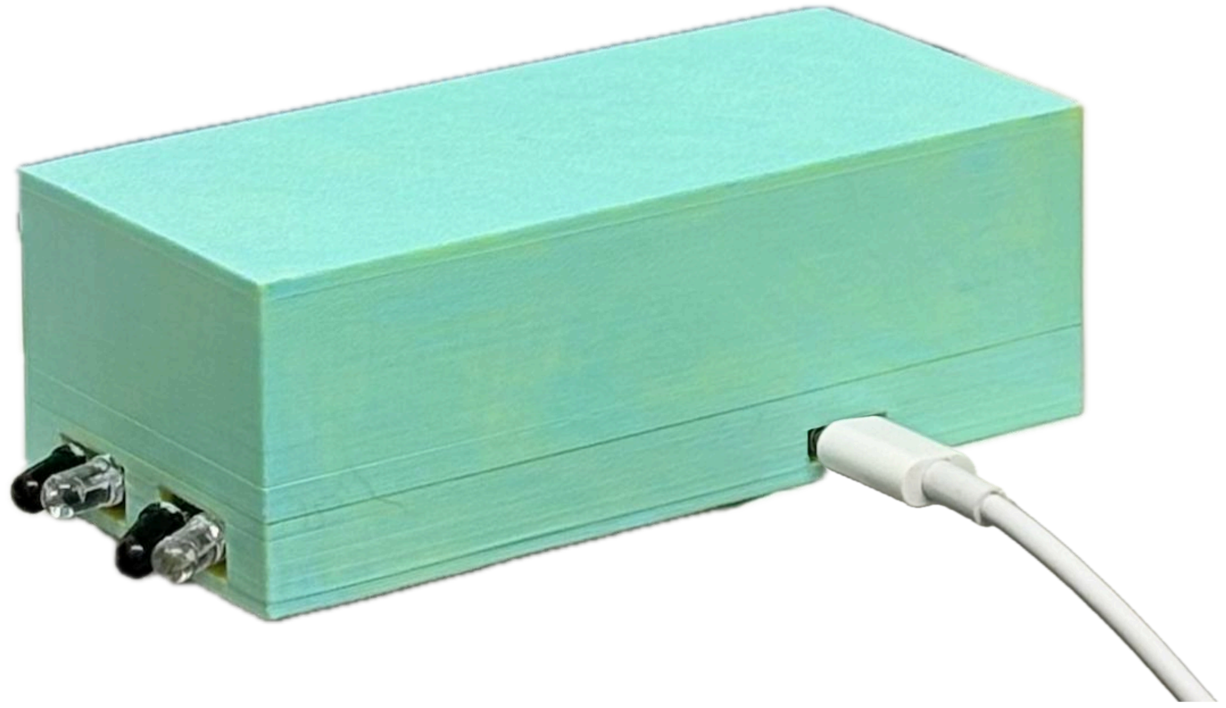
The updated count is sent via Bluetooth to the display device.

The display converts the count into a needle position + LED indicators.

Accuracy: ~95% correct detection during hallway testing with multiple users.

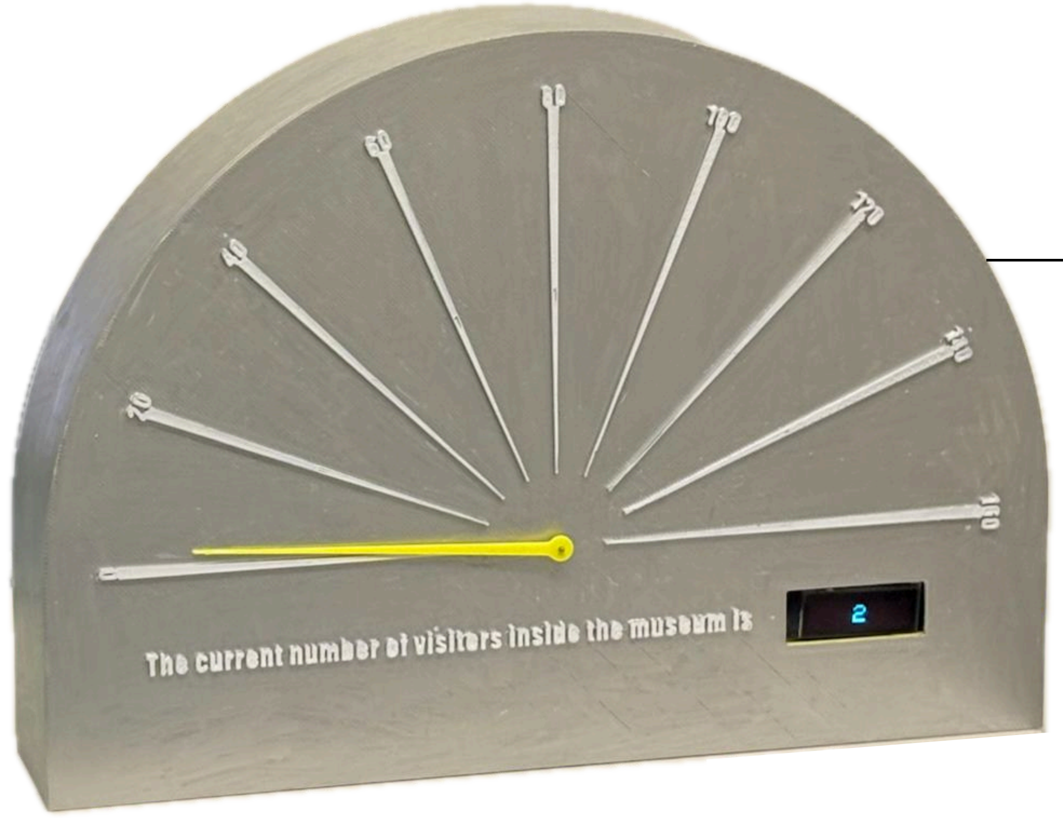
Battery considerations

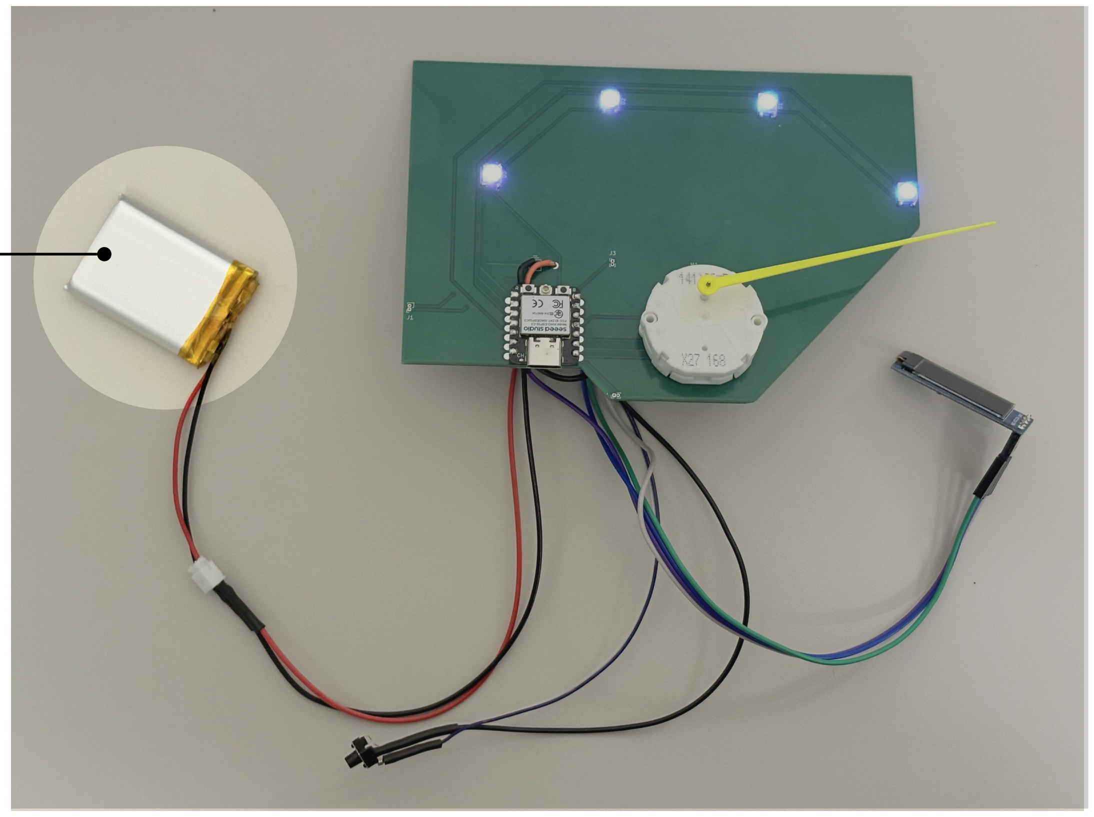
Sensor Device：USB-C wired power



Permanently installed at the museum entrance, so UCB-C wired power is practical and reliable. No battery drain concerns and can run 24/7 without maintenance.

Display Device Power: LiPo battery (1200mAh)





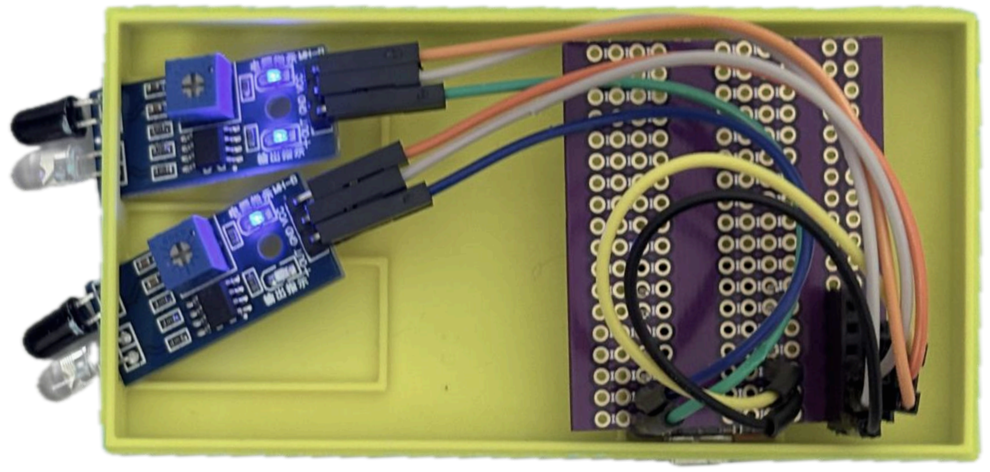
Placed on a staff desk or mounted on a wall. So Battery used to be portable and cable-free for a clean presentation.

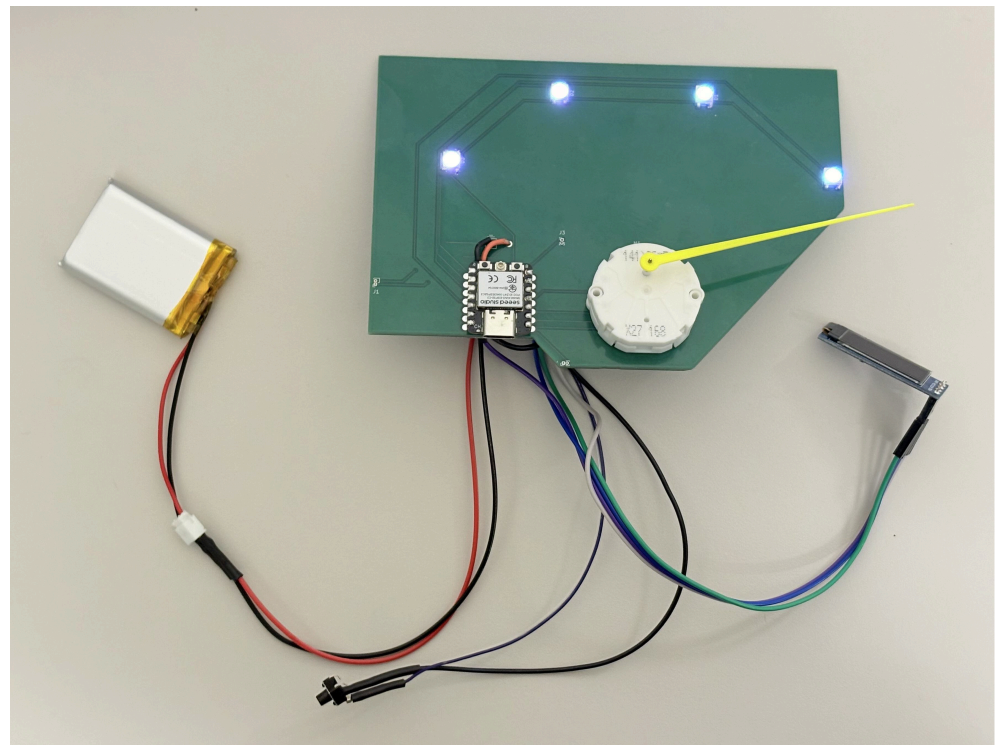
Display Device Power Estimate:

Component	Typical Current
XIAO ESP32-C3 (BLE active)	~50mA
SSD1306 OLED (128x32)	~20mA
SwitecX25 stepper motor (moving)	~30mA
4× NeoPixels (white, brightness 200)	~40mA
Total	**~140mA**

Budget summary

Component	Qty	Unit Price	Total
Seeed XIAO ESP32-C3	2	$4.99	$9.98
IR Sensor	2	$3.95	$7.90
SSD1306 OLED 128×32 (0.91")	1	$6.99	$6.99
Switec X25 Stepper Motor	1	$8.50	$8.50
NeoPixel LED	4	$0.50	$2.00
Push Button	1	$0.50	$0.50
LiPo Battery 1200mAh	1	$9.95	$9.95
Custom PCB	1	$5.00	$5.00
Total			$50.82





Future work

1

WiFi dashboard

Add WiFi connectivity to push real-time occupancy data to a cloud dashboard for remote monitoring

2

Multi-entrance support

Deploy multiple sensor nodes at different entrances, aggregating counts on one central display

3

Automatic HVAC/lighting integration

Use occupancy thresholds to trigger smart building controls (lights, air conditioning) via MQTT or Home Assistant

4

Weatherproof enclosure

Design a 3D-printed enclosure for outdoor or high-traffic installation